

23rd Feb 2026 | DA6401

Vanilla GD =

Batch GD

⇒ Momentum-based GD

- Problems of VGD -

1. slower convergence in Ravines

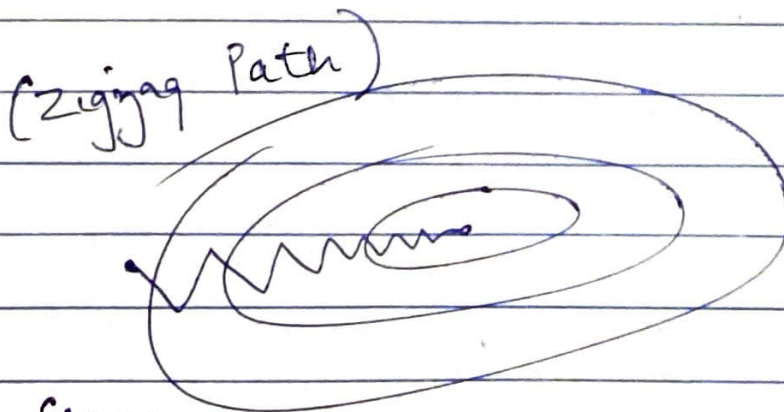
2. oscillations around minima

3. no possibility of acceleration mechanism

- Ravines - areas where surface curves more steeply in 1 dim. from another
→ other name 'Saddle'

→ each update is independent as there is no memory of previous direction. Update at the current step (magnitude) is solely dependent on the current gradient only. ($\theta_{t+1} = \theta_t - \eta \nabla_{\theta} L(\theta_t)$)

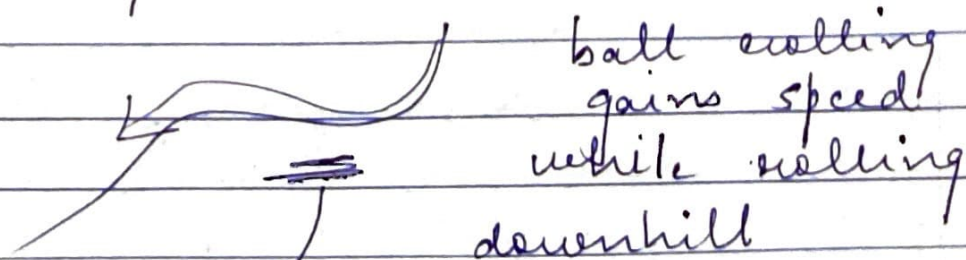
- Key issue → no way to accumulate velocity in fav. conditions or dampen the oscillations in unfav. ones.



Slow

progress, many oscillations, wasting many steps

- General Intuition for Momentum
Ball rolling down a hill



even-though gradient is 0,
it keeps moving.
Maintains a velocity vector.

- earlier, $\theta_{t+1} = \theta_t - \eta \nabla_{\theta} L(\theta_t)$

Momentum-based

$$v_{t+1} = \gamma v_t + \eta \nabla_{\theta} L(\theta_t)$$

$$\theta_{t+1} = \theta_t - v_{t+1}$$

update depends on accumulated velocity v_t . $\gamma (0.9)$ controls how much past.

- Update rule

$$v_0 = 0$$

t iteratⁿ

$$\left[\begin{array}{l} v_{t+1} = \gamma v_t + \eta \nabla_{\theta} L(\theta_t) \\ \theta_{t+1} = \theta_t - v_{t+1} \end{array} \right]$$

$$\gamma \in [0, 1)$$

$$\eta > 0, \quad v_t = \text{velocity (accumulated gradient)}$$

$\gamma = 0$ (vanilla GD)

$\gamma = 1$ (max. momentum, very smooth trajectory)

- EMA (Exponential Moving Avg.)

v_t is EMA of all past gradients

$$v_t = \eta \sum_{i=0}^{t-1} \gamma^{t-1-i} \nabla_{\theta} L(\theta_t)$$

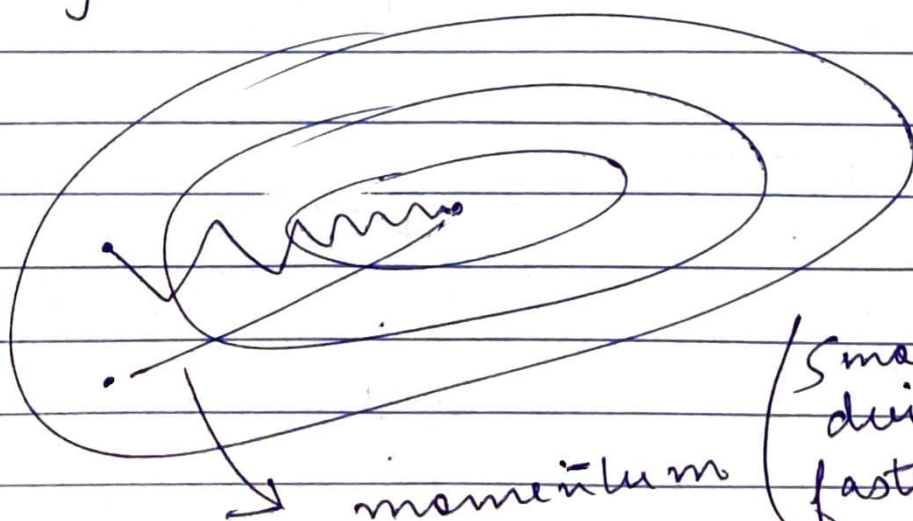
recursion is coming from

$$v_{t+1} = \gamma v_t + \eta \nabla_{\theta} L(\theta_t)$$

- Weighted Sum of Gradients

$$v_t = \eta (g_{t+1} + \gamma g_{t-2} + \gamma^2 g_{t-3})$$

recent gradient have more influence $\gamma^0 = 1$. They decay exponentially with age.



trajectory
(smooth, direct path, faster convergence)

- when to use $\rightarrow$ always recommended for DNN. Particularly efficient for high-dim. optimizaⁿ, noisy gradients, ill-conditioned loss surfaces.
- this converges faster than ∇ GD. Both stochastic & mini-batch of momentum GD exist.

$\Rightarrow$ Nesterov Acc. GD

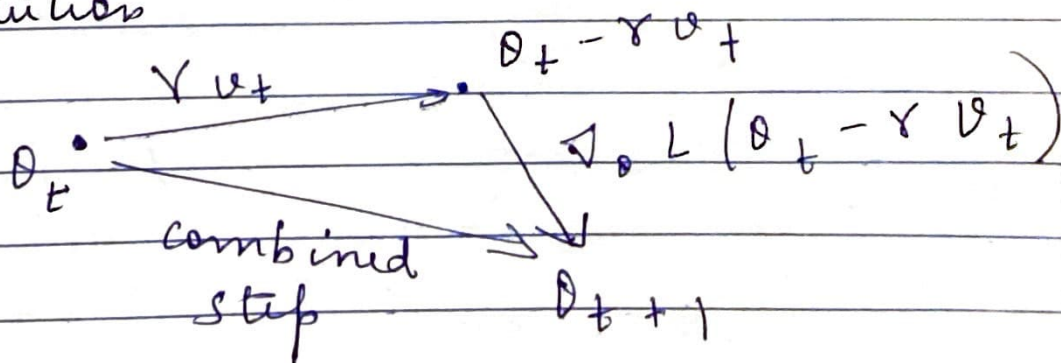
- limita^{ns} of momentum-based
 1. excessive overshooting
 2. unnecessary oscilla^{ns} (U-turns)
 3. slower convergence near the minima
 4. multiple small correc^{ns} is needed.

$$v_{t+1} = \gamma v_t + \eta \nabla_{\theta} L(\theta_t)$$

evaluated at current posⁿ

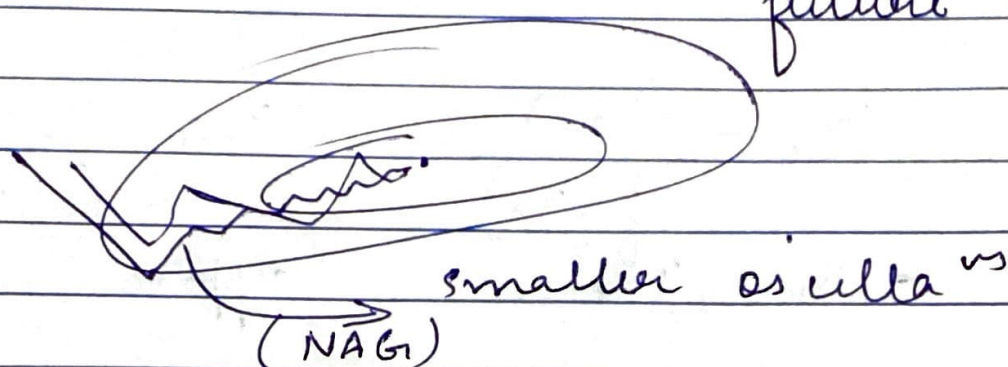
no look-ahead mechanism.

- intuition



→ first take a step. Compute gradient at look-ahead posⁿ. Make the app. correcⁿ.

- NAG leads to fewer oscillaⁿ, smoother convergence, better performance near the minimum. We try to anticipate future in here.



- Update rule -

$$\begin{cases} v_{t+1} = \gamma v_t + \eta \nabla_{\theta} L(\theta_t - \gamma v_t) \\ \theta_{t+1} = \theta_t - v_{t+1} \end{cases}$$

$\gamma = [0, 1)$ is momentum coeff.
 $\eta > 0$ is learning rate.

look-ahead posⁿ

$\begin{cases} \nabla_{\theta} L(\theta_t) & \rightarrow \text{Momentum based} \\ \nabla_{\theta} L(\theta_t - \gamma v_t) & \rightarrow \text{NAG} \end{cases}$

key difference